Client Evaluation

1. **Answered Problem Sections**
   4.5/5. It covers the whole question I posed, from cleaning all the way to a final flagged list of intersections with reasons attached, so an engineer could actually act on it. I'm taking one point off because the reported metrics in the writeup don't always match what the code prints, so a few of the answers are hard to fully trust as written. For example:
   In the Section 2.4:

   ```
   Baseline MAE:  0.812
   Baseline RMSE: 1.748
   ```

   Since the client cares about the top 30:

   However the markdown said:

   The baseline achieved:

   - **MAE:** 0.624 casualty crashes
   - **RMSE:** 1.366 casualty crashes
   - **Top-30 overlap:** 14/30 intersections (46.7%)

   So I'm not quite sure how this is calculated.
2. **Technique judgment**
   5/5. She chose supervised regression and splitting by year for a forecasting task and 2022 never leaks into training which is all good.

3. **Evaluation/interpretation quality**
   4/5. The metrics do connect back to the real goal. However, as mentioned above, some markdown didn't correctly capture the code. The baseline text says MAE 0.624 and 14/30 but the code says 0.812 and 16/30.
   Also in the Session 2.2

```
intersections["Target_Casualty_Crashes_2022"].describe()
```

```
count    5337.000000
mean        1.297358
std         3.453294
min         0.000000
25%         0.000000
50%         0.000000
75%         1.000000
max        58.000000
Name: Target_Casualty_Crashes_2022, dtype: float64
```

But the markdown said:

> There are 8,755 intersections, but 6,243 (71.3%) had zero casualty crashes in 2022. The mean is only 0.78 while the maximum is 34, so the target is extremely right-skewed and zero-heavy.

I dont see the match in any of the numbers here.

4. **Communication**
   5/5. Very clean. The summary did a nice job of using plain English to explain the concept for the client.

Open ended feedback
Overall this is a very solid work and the methods used are good. The year based split, using the trees to show how stable the ranking is and being upfront that the models only just beat the baseline all show good judgement. She is not overselling the results. My main worry is that the written markdown look like they didn't come from the code or maybe it is from older run (?) so in a few spots the text and the actual outputs don't match.